

Lab 1 – TrashTag Product Description

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1 Introduction

Litter and illegal dumping is a common problem all people have seen. Litter is not just unsightly; it is dangerous and directly impacts the environment and all of those living in it, including humans. Improperly discarded waste accounts for an estimated 100,000 marine animal deaths annually (Texas Disposal Systems 2024). According to a survey from 2016 roughly 21% of beachgoers report they have been injured from litter such as glass or metal from bottles and cans. The danger and ugliness of litter drives people away. The City of Hampton reports a drop in property values of 7-10% in areas with illegal dumping. Tourist destinations with poor litter control and cleaning see drops in revenue up to 38% (Sorensen 2024). Furthermore, when people see litter, they are more likely to add to it instead of finding proper disposal locations (Blouin 2016).

Keeping public areas clean is constant effort. Volunteers in all states of the US have always cleaned up litter on their own. In just the state of Virginia in the last 36 years an approximate 7.1 million pounds of litter have been found and properly disposed of (Chesapeake Bay Foundation 2025). It can take hundreds of such volunteers to scour the miles and miles of land for any given public recreation area. Those volunteers are still not enough. Local governments still end up paying for litter collection, removal, and disposal. In 2020, municipalities across Texas spent over \$50 million on trash remediation (Sorensen 2024).

When small crews or individual volunteers come across large, dumped items such as appliances, mattresses, or tires, they frequently must leave them behind so that a larger team or properly equipped people can return to haul them away. Many times, individuals that take notice of such bulk items either do not know the proper authority to which to report the trash or they forget by the time they leave the area where they saw the problem. These items can then languish there for months or years contributing to all the problems above.

Illegal dumping is a problem that requires multiple solutions. The reason people dump large items in inappropriate places is nearly always a convenience factor (Blouin 2016). This needs to be solved with expensive buy-in to solutions sponsored by local governments, or in the case of federal land, the federal government. Such programs can be extremely difficult to start due to the bureaucracy involved and difficult to maintain due to budget limits, cuts, and reallocations, especially for very small municipalities in rural areas.

The gap becomes clear between those that see and are affected by illegal dumping and those with the means and motivation to remove it and dispose of it properly. Giving both sides of this equation an efficient means of communicating can close this gap leading to a cleaner and safer environment for people, plants, and animals. TrashTag aims to be this solution by giving citizens the ability to take a photo of the problem they find and easily upload it to a map that anyone can see, including local governments and environment clean-up organizations. At a glance people and organizations can see what issues are nearby to them and effortlessly gauge the scope of the problem to plan accordingly. Combine this with tools that let citizens organize themselves and provide local information about proper bulk item disposal, citizens are empowered to make a significant change in their local area thus making a global impact.

TrashTag also includes additional features to further support the clean-up efforts. Users are prompted when nearby previously tagged trash so that they can verify it, both for being there or if another user has already cleaned it. The system is also gamified for further incentives where users can track their impact and compare their progress with other local users. Finally, organizations and local governments can get access to analytical data showing hotspots of dumping and track the efforts of clean up over time. Empowering individuals and groups, letting them all connect and plan makes this large work easier for everyone.

2 Product Description

TrashTag aims to improve the litter and waste cleanup process by developing a mobile application where users can photograph and report piles of waste, give their exact locations, and connect with cleanup groups that are well-equipped to handle the cleanup process in order to improve environmental wellbeing and preserve nature's beauty for the public eye.

TrashTag is committed to helping the environment and the public by providing a resource that clearly highlights the location, extent, type and weight of the waste as well as provides updates regarding existing cleanup efforts, dangers and geographic littering patterns to help cleanup crews allocate their time and resources more effectively.

2.1 Key Product Features and Capabilities

The core functionality of TrashTag is a live map of reported bulk litter and dumping. Users can report litter and attach photos to the report. The map lets users see which litter is nearby to their location or see reports in an area of interest. Those reports show as items on the map for other users who can then see the full details. The images on the reports allow for clean up crews to accurately gauge the scope of the litter and plan accordingly.

Users can indicate on reports if the trash is still there or if they have cleaned it. Which users have reported and which users have cleaned is stored. TrashTag uses that data to maintain a ranking system for users based on how much they have reported and cleaned. Users can also update their own reports to add additional information.

When TrashTag detects a user is nearby reported trash, they are given the option of verifying that it still is there. If a user has indicated they have cleaned previously reported trash, nearby users are requested to verify the trash is gone.

Users can also organize their own clean-up efforts, targeting a specific reported trash item or a series of reports. Users can then schedule these meet-ups and communicate directly with one another.

2.2 Major Components (Hardware/Software)

TrashTag has big goals, but the implementation is relatively simple. The client front end is built in React and provides the map interface using OpenStreetMap. This is a web app designed to display on mobile or desktop easily thus making native mobile apps unnecessary. The back end is handled with Python and Django. For the prototype, the data is stored in a basic SQLite database. However, through the power of object orientation and the Django Object Relational Model, the database can easily be switched with a more powerful one such as MySQL. Digital Ocean droplets provide powerful, flexible, and affordable co-hosted virtual machines that can be expanded easily to handle additional users. **Figure 1** *Major Functional Components Diagram* (below) shows how each of these pieces related and connect to one another.

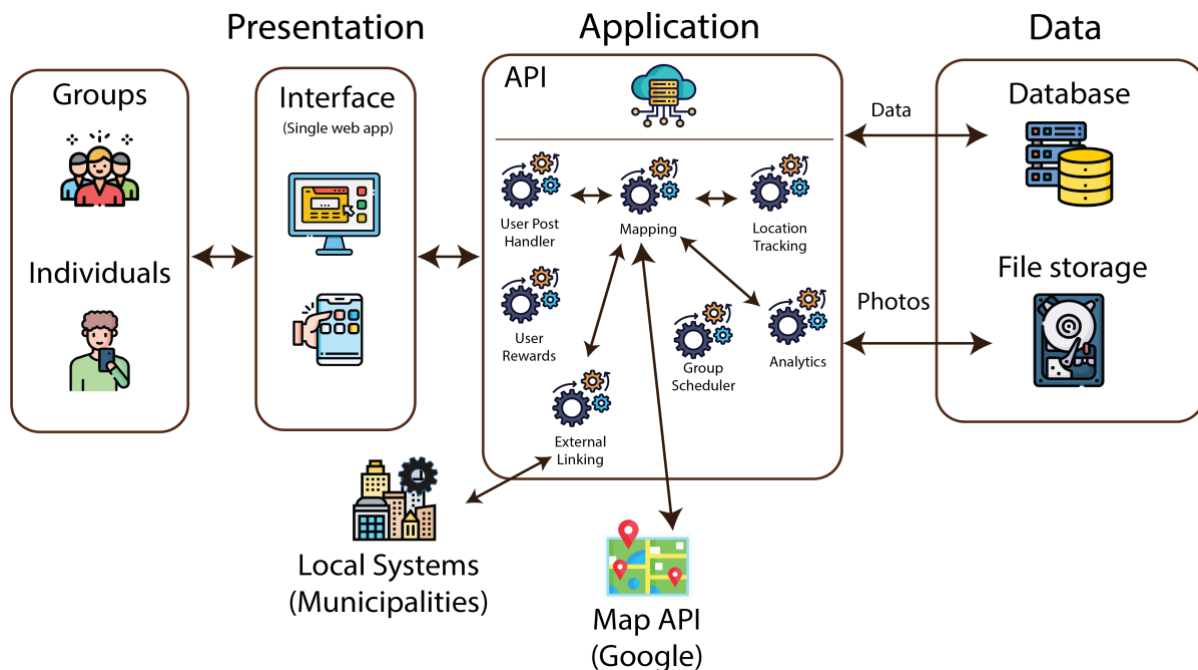


Figure 1
Major Functional Components Diagram

3 Identification of Case Study

The local non-profit organization, San Marcos River Foundation (SMRF), is the focus of the first case study. SMRF is dedicated to the clean-up of the San Marcos River in San Marcos, Texas. They recruit and manage volunteers to perform clean-up events. SMRF struggles to efficiently use these volunteers as there are dozens of miles of river to search for trash, litter, and dumping. Knowing where dumped items are located lets SMRF use their volunteers much more effectively and quickly. The features SMRF would make most use of are the live map of trash, the ability to mark trash as cleaned, organize clean-ups, and see analytics of trash hotspots.

The next case study focuses on the everyday citizen utilizing public spaces such as the San Marcos River. Frequently, a river-goer notices large trash or dumping and is unable to handle cleaning it. They lack the time, ability, knowledge, or equipment needed to move and safely dispose of the discarded objects. Ecologically conscious citizens rarely know to whom to report dumping, and even the most enthusiastic citizens can forget about what they saw by the time they are somewhere to look up the necessary information. The features these citizens would use the most are the live map of trash, photo upload report of found trash, and the verification of existing or cleaned up trash.

In both cases, TrashTag serves to quickly, easily, and efficiently communicate to each group about the trash in their community. As everyday citizens see the unsightly and dangerous dumped items, they can tell groups like the SMRF about them without needing to search for clean-up organizations or have phone calls or emails that go lost and unanswered. SMRF and similar organizations can confidently dispatch volunteers with the right equipment the first time. Volunteers feel more empowered and are more likely to volunteer again. Local governments can track citizen and non-profit effort and effectiveness to better gauge how taxpayer money can be dispersed.

A typical lifecycle of a single trash report looks like the following. A recreational river-goer walks down to the water's edge to enjoy a swim. Before they enter the water, they see that someone has illegally dumped a broken washing machine. They are not sure who to contact and do not have the ability or equipment necessary to haul it off themselves. They can use their phone to create a new report in TrashTag with a single button tap. They take a picture and enter in about how much they think it weighs. After tapping "Done", they can enjoy the rest of their day. Another swimmer later sees the same washing machine but also sees on the map that it has already been reported. They tap the report and tap "Verify". Later, someone from the San Marcos River Foundation views the map of their area and sees a new item on the map and that someone has verified its there. SMRF sees that it's a complete washing machine and knows they will need a truck and at least 3-4 people to get it out of the water and into the truck. SMRF is already familiar with where to properly dispose of such items. SMRF can then create an event asking for four volunteers to come to the river next weekend. Some locals see the event and sign up. They all arrive at the river at the event time, haul the washing machine into the truck, and off to the recycling center it goes!

4 Glossary

- **Analytics:** The collection of data sent to local government authorities and/or organizations providing them with snapshots of data regarding littering in specified areas.
- **API:** Application Programming Interface. A software mechanism of accepting and providing data from and to external applications.
- **Dumping/Trash/Litter:** Any material purposefully or otherwise left in an area where it does not belong.
- **Environmental Organization:** Private, and frequently non-profit, groups dedicated to environmental causes.
- **Geotag/Geotagging:** GPS provided location information embedded into images taken with a camera.
- **Report/Reporting:** The set of information and photos that users individually enter in the application to indicate illegally dumped trash they have found in the public.

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